INT404 Lab 4 - Incident Response & Forensics

Course: Threat Analysis & Mitigation Proposal

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Date: 05/04/2026

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Threat Analysis & Mitigation Proposal

Part 1: Threat Analysis Report

1. Critical Infrastructure Sector: Food and Agriculture

The Food and Agriculture sector is a vital component of national and global critical infrastructure. It encompasses food production, processing, distribution, and retail systems that ensure the availability of safe and reliable food supplies. Disruptions in this sector can have severe consequences, including food shortages, economic instability, and threats to public health.

Modern food production systems rely heavily on digital technologies such as automated processing systems, supply chain management platforms, and Industrial Control Systems (ICS). While these technologies improve efficiency, they also introduce significant cybersecurity risks.

A major real-world example highlighting these risks is the **JBS Foods Ransomware Attack 2021**, one of the largest cyberattacks against the food supply chain.

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2. Overview of the Case Study

JBS Foods, the world's largest meat processing company, was targeted by a ransomware attack in May 2021. The attack forced the company to shut down operations across multiple countries, including the United States, Canada, and Australia.

The disruption affected meat production significantly, temporarily reducing supply and raising concerns about food shortages and price increases. The company ultimately paid an $11 million ransom to regain access to its systems.

This incident demonstrates how cyberattacks can directly impact critical infrastructure and everyday life.

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3. Primary Cyber Threats

The **JBS Foods Ransomware Attack 2021** illustrates several key cyber threats affecting the Food and Agriculture sector:

a. Ransomware Attacks

Ransomware is the most prominent threat in this case. Attackers encrypted JBS's systems, forcing the company to halt operations. This type of attack is particularly dangerous in industries that rely on continuous production.

b. Supply Chain Disruption

The attack disrupted not only JBS but also suppliers, distributors, and retailers. Cyberattacks in this sector can cascade across the entire food supply chain.

c. Operational Technology (OT) Targeting

Food processing facilities use automated machinery controlled by digital systems. Attacks targeting these systems can shut down production lines entirely.

d. Data Exfiltration and Extortion

Modern ransomware attacks often involve stealing sensitive data before encryption. This allows attackers to threaten public release of data if the ransom is not paid.

e. Phishing and Initial Access Attacks

Although exact entry methods are not always disclosed, many ransomware attacks begin with phishing emails or compromised credentials.

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4. Key Vulnerabilities

The incident reveals several vulnerabilities common in the Food and Agriculture sector:

a. Lack of Network Segmentation

Poor separation between IT and OT systems allows attackers to move laterally and impact production systems.

b. Weak Access Controls

Inadequate authentication mechanisms, such as lack of multi-factor authentication (MFA), make it easier for attackers to gain access.

c. Legacy Systems

Many food production facilities rely on outdated systems that are difficult to secure and patch.

d. Limited Cybersecurity Investment

Compared to sectors like finance or energy, food companies often invest less in cybersecurity, increasing their risk exposure.

e. Insufficient Monitoring and Detection

Delayed detection allows attackers to remain in systems longer, increasing damage.

f. Third-Party Risks

Vendors and partners may introduce vulnerabilities if their security practices are weak.

Part 2: Mitigation Proposal

Organization: JBS Foods (Post-Attack Security Improvement Plan)

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1. Scope

This cybersecurity assessment will include:

- Corporate IT systems (email, databases, enterprise systems)
- Operational Technology (OT) systems used in food processing plants
- Network infrastructure (firewalls, routers, VPNs)
- Supply chain and third-party integrations
- Employee access controls and cybersecurity awareness

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2. Objectives

The primary objectives are:

- Identify vulnerabilities across IT and OT environments
- Strengthen defenses against ransomware attacks
- Improve detection and response capabilities
- Enhance supply chain security
- Protect sensitive business and operational data
- Ensure compliance with cybersecurity best practices

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3. Methodologies

a. Vulnerability Assessment

Automated tools will scan systems for known vulnerabilities, outdated software, and misconfigurations.

b. Penetration Testing

Ethical hacking techniques will simulate real-world attacks, including:

- Network penetration testing
- Internal system exploitation
- Ransomware attack simulations

c. Phishing Simulation

Employees will undergo simulated phishing attacks to test awareness and reduce human-related risks.

d. Network Segmentation Testing

Testing will ensure proper separation between IT and OT systems to prevent lateral movement.

e. Security Configuration Review

Assessment of:

- Access controls
- Encryption protocols
- Firewall configurations

f. Incident Response Testing

Tabletop exercises will simulate ransomware scenarios to evaluate response effectiveness.

g. Backup and Recovery Testing

Verification of secure, offline backups and recovery procedures to ensure business continuity.

h. Third-Party Risk Assessment

Evaluation of vendor security practices to reduce supply chain risks.

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4. Recommended Mitigation Strategies

Based on the findings, the following measures are recommended:

- Implement **multi-factor authentication (MFA)** across all systems
- Deploy **advanced endpoint detection and response (EDR)** tools
- Strengthen **network segmentation** between IT and OT environments
- Maintain **regular offline backups**
- Conduct ongoing **employee cybersecurity training**
- Establish a robust **incident response plan**
- Perform regular **security audits and updates**

Conclusion

The **JBS Foods Ransomware Attack 2021** demonstrates the significant cybersecurity risks facing the Food and Agriculture sector. As digital transformation continues, cyber threats such as ransomware pose increasing risks to food supply chains and global stability.

By implementing comprehensive security measures and proactive risk management strategies, organizations like JBS Foods can enhance resilience, protect critical operations, and ensure the continued availability of essential resources.

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